

Additional Simple Linear Regression Practice Question:

BrightRoof SA is a national residential solar-installation company. During one quarter, the company ran separate paid-search campaigns in 72 sales areas across South Africa. Management wants to determine whether higher local Google Ads expenditure is associated with higher values of signed solar installation contracts.

Each observation represents one sales area.

- Dependent variable: value of signed installation contracts, in thousands of rand.
- Independent variable: Google Ads expenditure, in thousands of rand.

A simple linear regression model was fitted, producing the following (redacted) output:

Coefficients:

	Estimate	Std. Error	t value	p-value
(Intercept)	210.6400	15.2000	13.86	---
google_ads_spend	18.7500	1.8500	10.14	---

Residual standard error: --- on 70 degrees of freedom

Multiple R-squared: ---

F-statistic: --- on 1 and 70 degrees of freedom

The following analysis-of-variance table was also constructed:

Source of variation	df	Sum of Squares	Mean Square	F
<i>Regression</i>	1	<i>A</i>	<i>A</i>	<i>C</i>
<i>Error</i>	70	126 437.50	<i>B</i>	
<i>Total</i>	<i>D</i>	311 977.50		

Unless stated otherwise, round final numerical answers to two decimal places and do not round during intermediate calculations.

Questions:

1. Write down the fitted simple linear regression equation. Interpret the estimated coefficient of Google Ads expenditure in the context of BrightRoof SA.
2. Calculate the values of *A* and *D* in the analysis-of-variance table.
3. Calculate the error mean square, *B*, and hence calculate the residual standard error. Interpret the residual standard error in its original rand units.
4. Calculate the overall *F*-statistic, *C*.
5. Use the *F*-statistic to test the overall significance of the fitted regression model at the 5% level of significance. Your answer must include:
 - a) the null and alternative hypotheses;
 - b) the test statistic and appropriate degrees of freedom;

- c) the statistical decision based on the p-value approach; and
 - d) a conclusion within the context of BrightRoof SA.
6. Calculate the coefficient of determination, R^2 , and interpret it in the context of this study.
 7. Calculate the sample Pearson correlation coefficient between Google Ads expenditure and the value of signed installation contracts. Briefly interpret its direction and strength.
 8. A second simple linear regression model is fitted to the same 72 sales areas, using local radio-advertising expenditure as the predictor. This model has a residual standard error of 47.90. Based only on the residual standard errors, which of the two models provides the better fit to the observed contract values? Explain briefly.
 9. Use the fitted regression equation to estimate the value of signed installation contracts for a sales area in which BrightRoof spent R12 000 on Google Ads. Give your answer in rand.